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ITI - AMS Layout Course - Lab 08

Chip Floorplan, Integration, and Verification

Outline

- Part1: Chip floorplan and power planning
- Part2: Creating a layout with pins
- Part3: Exporting LEF and DEF files and importing LEF to an abstract view

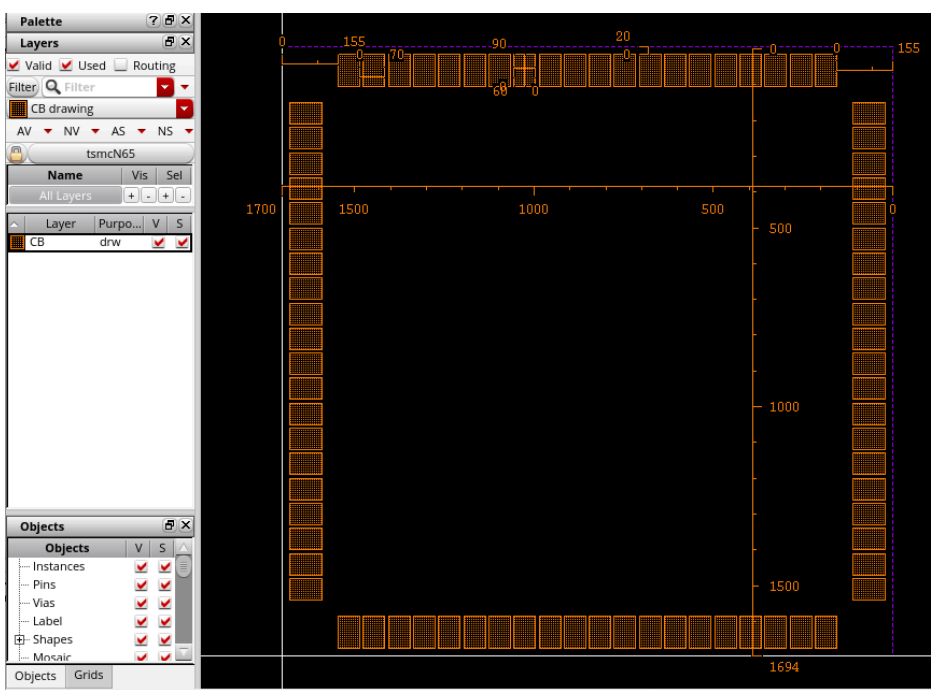
Objective

1. To be familiar with chip floorplan and power plan
2. To be familiar with LEF file and abstract view

Part1: Chip floorplan and power planning

Floorplan

1. Assume there is a chip with the following dimensions
    a. Chip size: 1.7mm x 1.7mm
    b. Pad size is 60um x 90um and its pitch is 70um
    c. Pad to chip edge spacing for the sealring is 20um
    d. Corner width is 155um.



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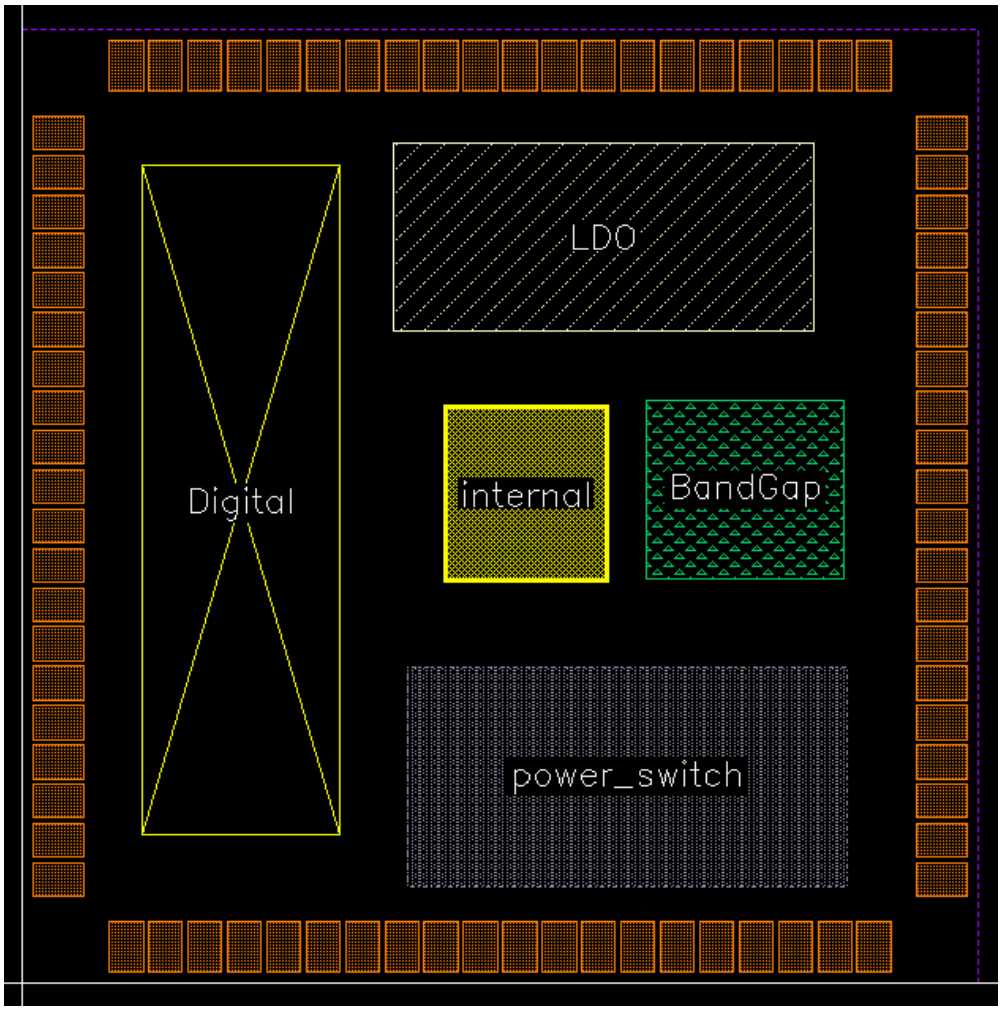
2. Assume that the chip has the following subsystems.
    a. Padframe
    b. LDO
    c. Bandgap
    d. Internal circuitry
    e. Power switch

3. Signal-driven & block-driven planning implies the following placement
    a. Assume the bandgap provides current to 3 blocks; LDO, power_switch, and internal. Therefore, the bandgap should be placed near them.
    b. Now, assume the LDO is carrying very high current in the pass device which is connected to the regulation output pad. Thus, the LDO should be placed near to its output pad to avoid routing high currents for long distances.
    c. Power switch is similar to the LDO. It needs to be placed close to its pads which are handling high currents.

4. Given that the blocks have certain given areas and aspect ratios, then the floorplan can be as shown below.



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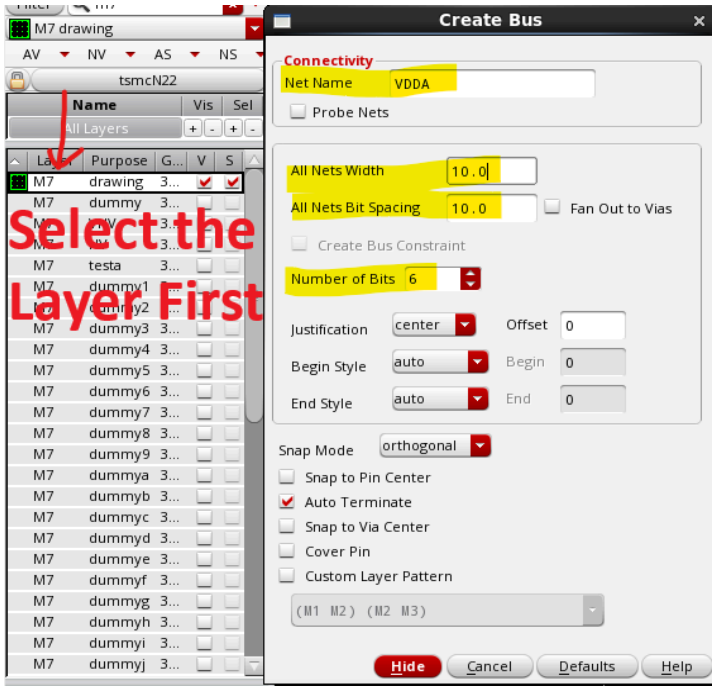
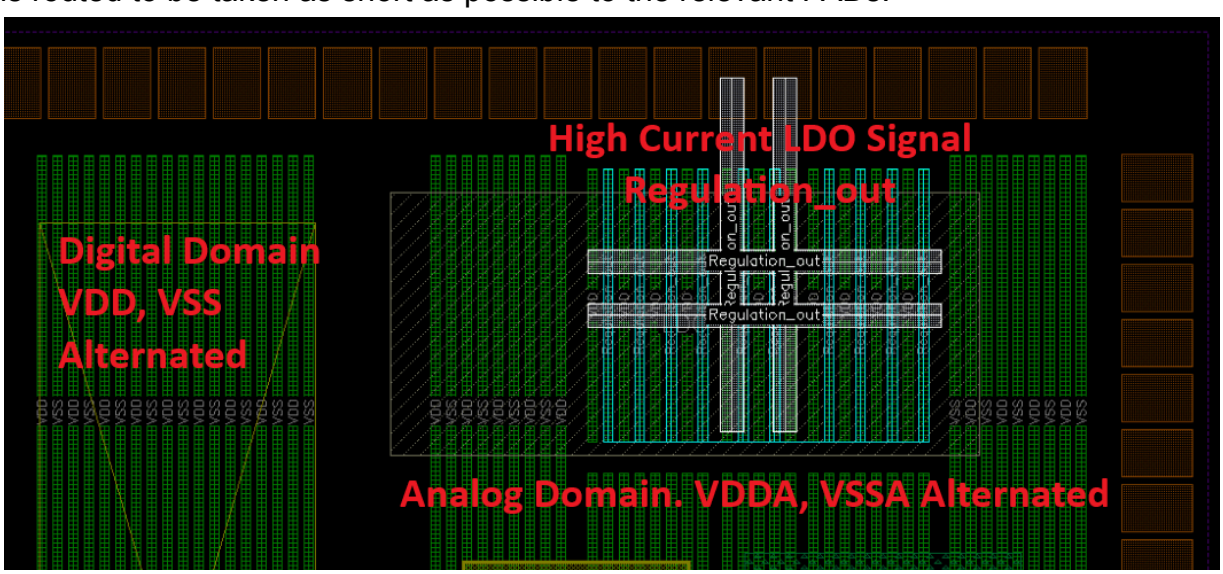
Power Plan

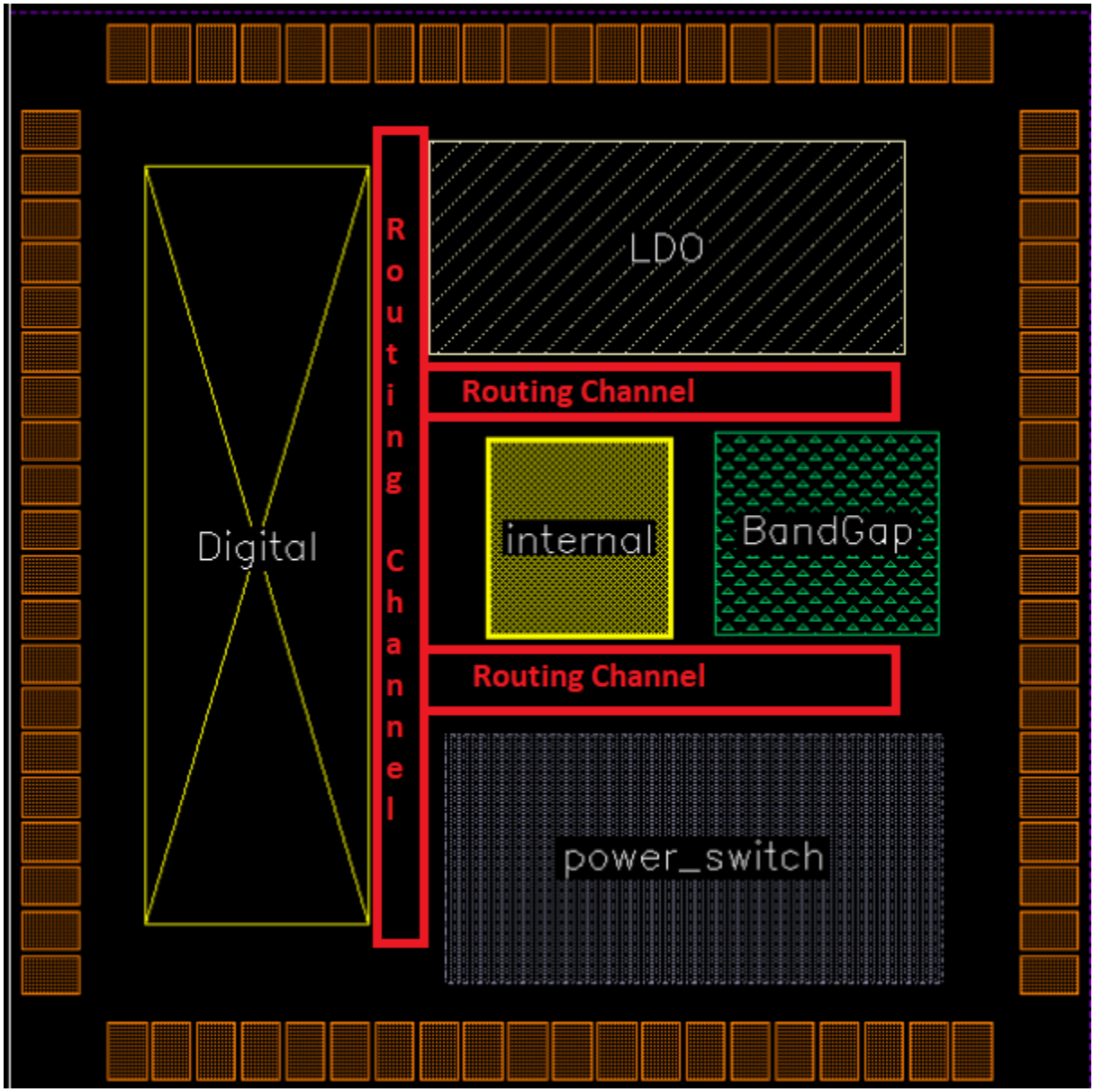
1. Press “Ctrl + Shift + x to call the bus routing which helps in power planning. Then press “**F3”** and choose the number of routes, their widths, and the spacing between them.



2. Plan the analog and digital domains supply routing, as well as sensitive and high current signals such as the regulated output of the LDO as shown below.
The digital supplies are routed over the digital block, while the analog supplies are routed everywhere else, and the LDO high current Regulation_output signal is routed to be taken as short as possible to the relevant PADs.



3. Plan your routing channels between the blocks. For example, the area highlighted in red between the digital and the remaining blocks on the right will be used to route the digital signals.



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Part2: Creating a layout with pins

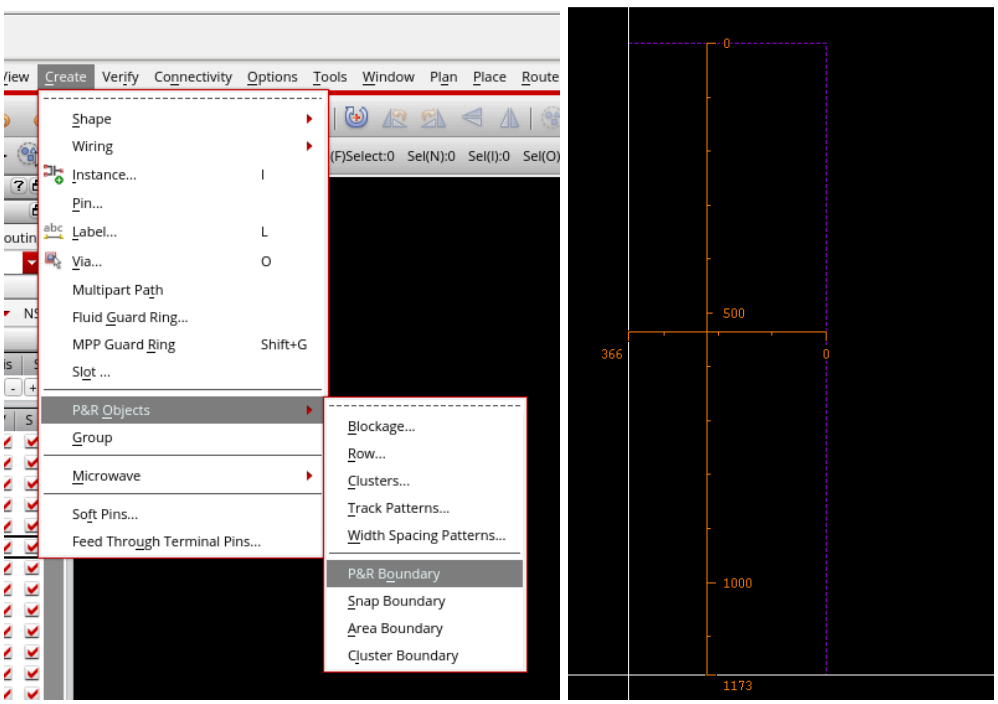
1. Assume there is a digital block that is going to be designed by the digital PnR team at the end of the project and we would like to connect to its interfaces from the early beginning to account for any issues as early as possible

2. The digital block has the i/o pins dig[0:9], VDD, and VSS

3. We will fix the pins location and send it to the PnR team to design based on these pin coordinates, so as to make it plug and play on the analog side

4. Hence it is decided to create a layout view with those pins which are defined as follows
    a. VDD is input & power signal
    b. VSS is input & ground signal
    c. Dig[0:9] are output signals

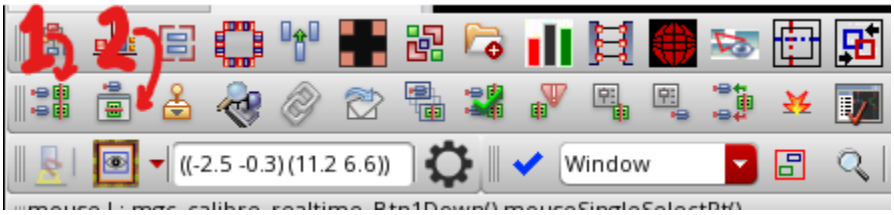
5. A “P&R Boundary” was created for the digital block given its aspect ratio as shown below



6. Usually the digital schematic is empty having pins only till the end of the project, so layout XL or SDL can be used to generate the pins in the layout.

7. Generating the pins:

a. The power and ground are on M5 and the signals are on M4.
b. Pins locations and dimensions are usually customized based on the givens and assumptions
c. There are two options to generate and distribute the pins.



i. **Generate all from source**
   - It gives you full control over the pins and its labels properties such as layers and dimensions, however all the pins are placed randomly beside each other.
   - This is a disadvantage with large blocks similar to the digital, you need to distribute the pins based on the block they are connected to. Hence, option 2 might be faster in such cases.
   - Here the pins distribution along the digital cell periphery can be done using Edit > Advanced > Align from the layout view.

ii. Generate **selected** from source.
   - This option gives the freedom to select some pins from the schematic, e.g. the pins going to the LDO, and place them in the layout view. Thus, you are planning the location of the pins while placing.
   - However, this option does not permit the user to edit the pins properties such as layers, dimensions which can be easily adjusted from Generate all from source.
   - So to adjust it, **open** “Generate all from source” first and select the required pins properties without completing the generation and this will be applied on the selected pins that you are placing in the layout.
   - **Example**
     Let’s assume we want to choose M4 pins and center-right label option as we are planning to place these pins on the right side of the digital block.The pins labels were adjusted to be the same as the pins with auto height (can be customized case by case)

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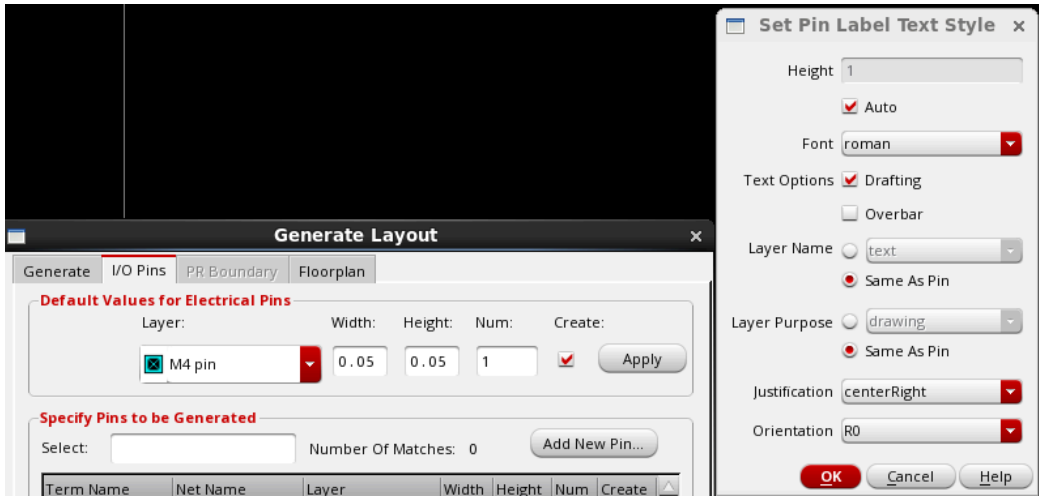
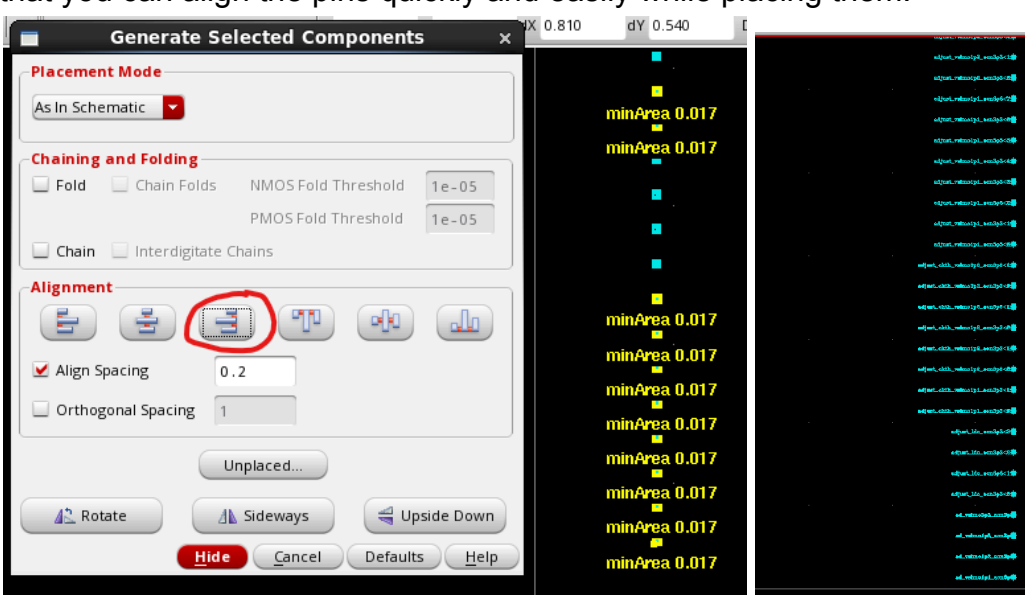


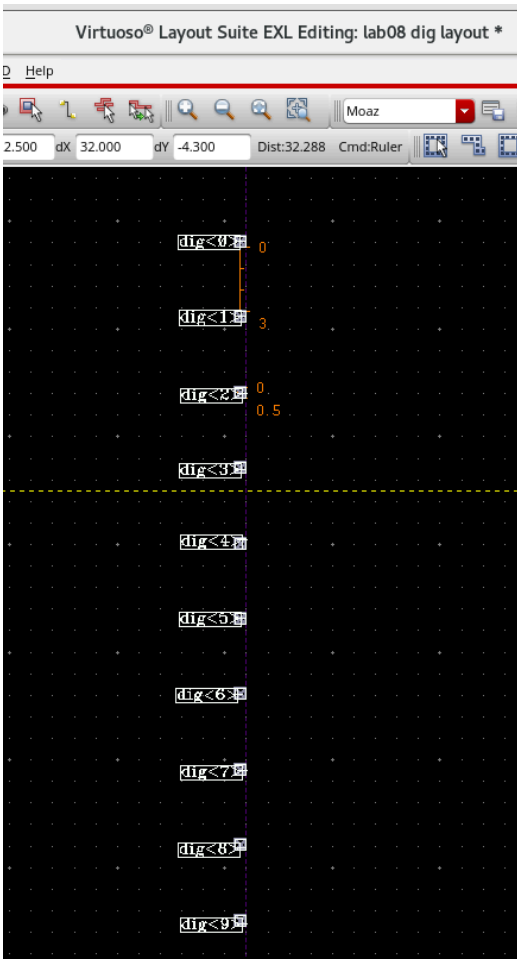
After adjusting the generate pins settings, cancel the “Generate all from source” which saves your pins preferences and then return back to the schematic, select the pins that you want to place in the layout view, and choose “Generate selected from source”.

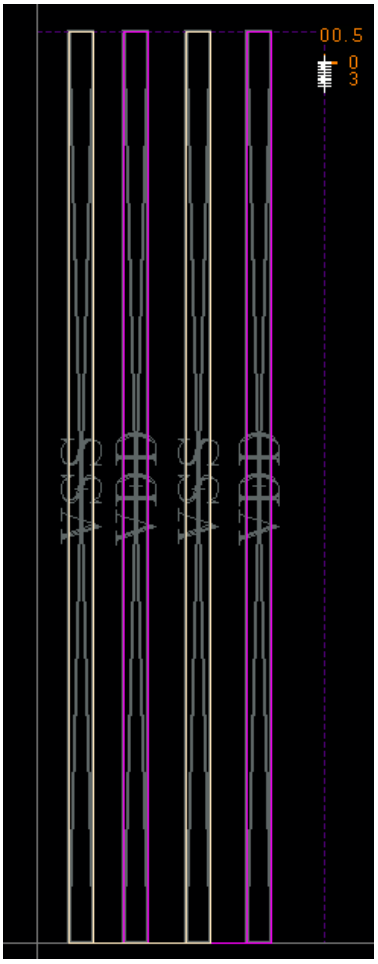
d. Assuming that we chose “Generate selected from source”, then associate it with pressing “F3” to show all the alignment options as shown below, so that you can align the pins quickly and easily while placing them.



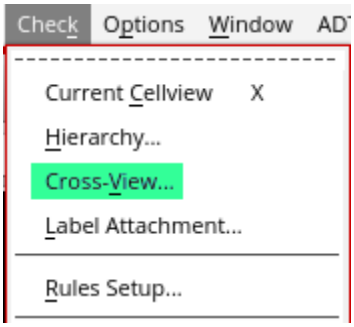
8. Adjust and finalize the pins’ locations manually
   a. Place the digital pins on the top right corner assuming this is the best location from the top level routing perspective.
   b. Use large widths for the power and ground and alternate them. Please note that sometimes the supply and ground are not included in the icon

view and left for the digital team to place it as a grid with calculated IR drops.

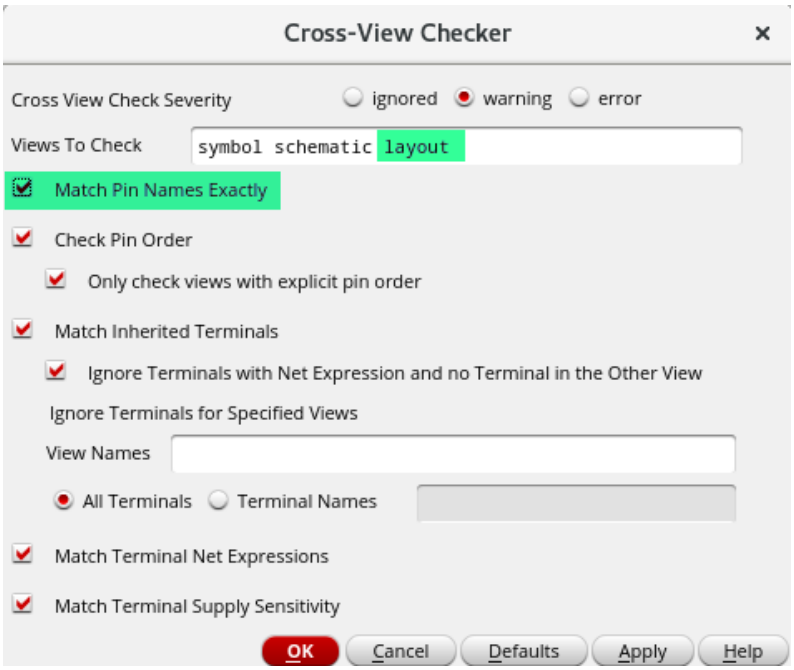




9. Now cross check between the schematic and the layout should be performed to ensure that pin names, and properties are the same.
   a. From the schematic view click on Check > Cross-View…



   b. Edit the views to check to be symbol, schematic, and layout



   c. Click on “Apply” and check the warnings in the CIW, some of the warnings can be fixed by scripts such as pins direction and others require manual intervention such as missing pins

```
Warning: Terminal "ib<7>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "ib<8>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "ib<25>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "ib<16>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "ib<6>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "ib<4>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "ib<17>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "ib<13>" in view schematic is (output) but is (input) in "BandGap layout".
Warning: Terminal "bg_ok" in view schematic is (output) but is (input) in "BandGap layout".
INFO (SCH-1172): There were 0 errors and 30 warnings found in "lab08 BandGap schematic".
```

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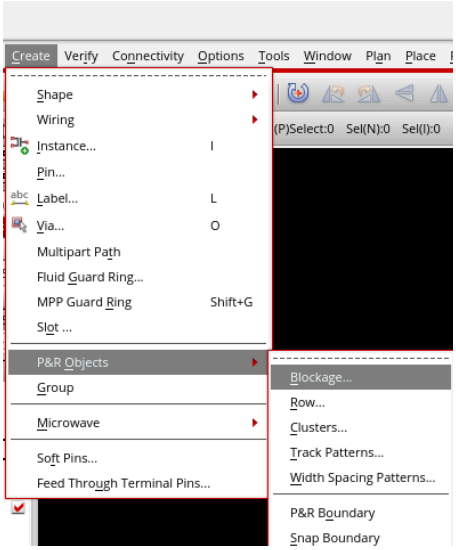
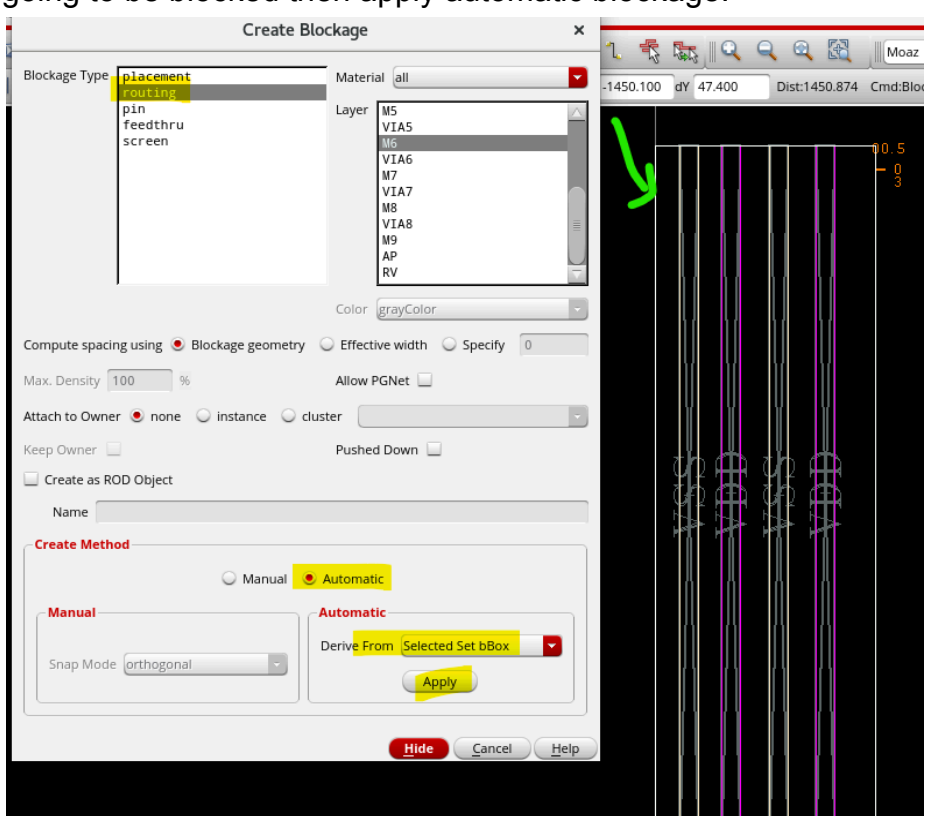
10. Go back to the layout and add Blockage layers for any layers above M5 to prevent the PnR tool from using them.
    a. Goto Create > P&R Objects > Blockage



    b. Select the PR boundary of the cell and in the opened window, select "Blockage Type" to be "routing" then choose the metal layers that are going to be blocked then apply automatic blockage.



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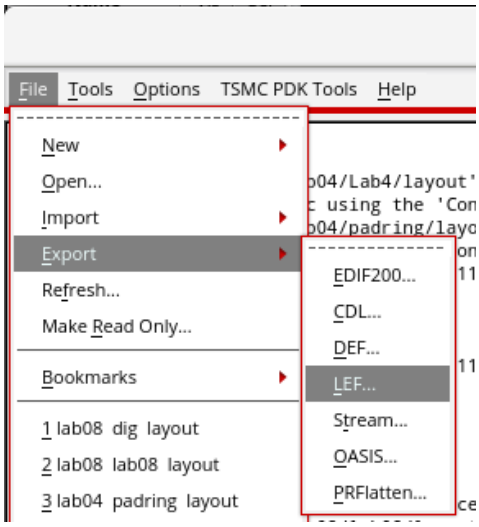
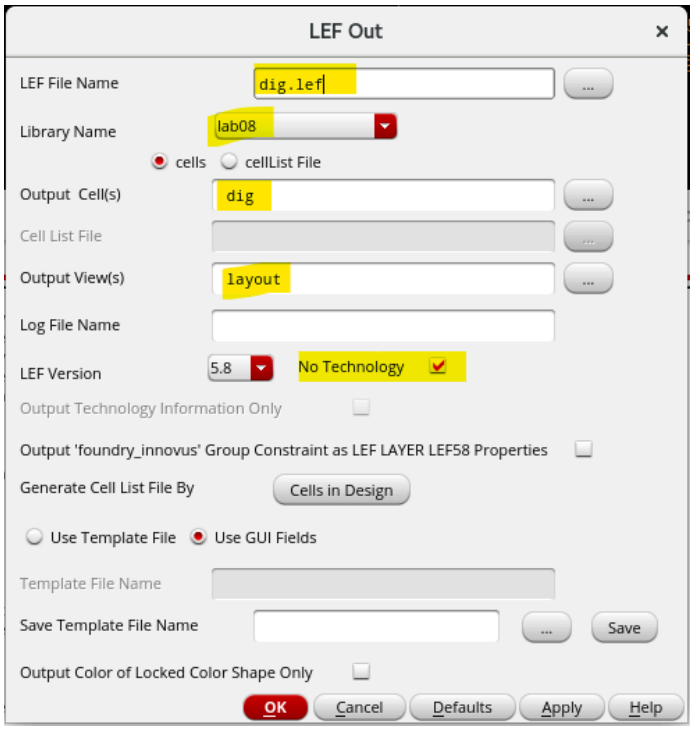
Part3: Export LEF and DEF files and import LEF as an abstract view

1. From the CIW choose “Export > LEF”

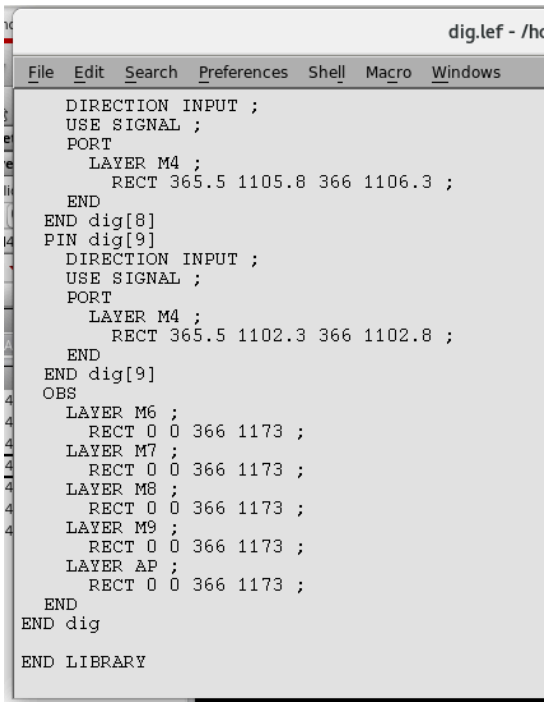


2. Enter the LEF file name, the cell name, and select “No technology”



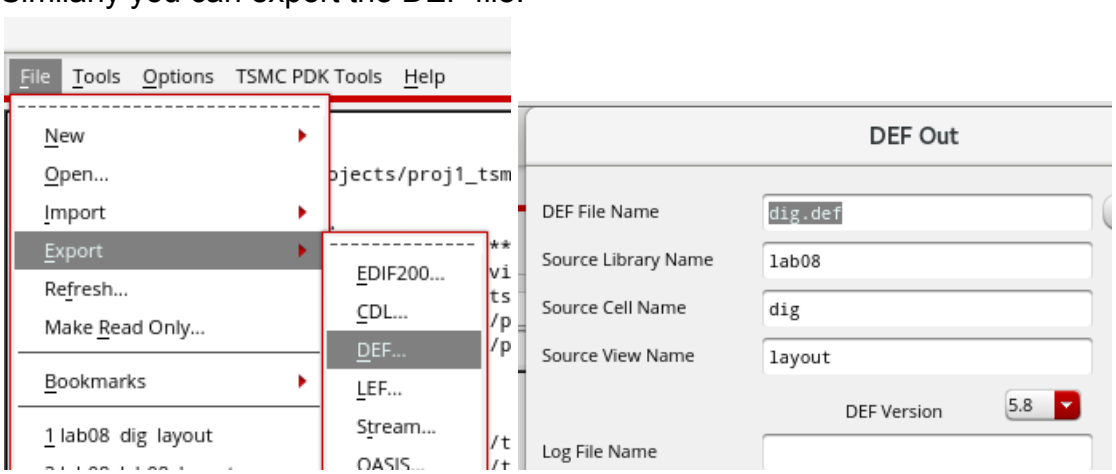
3. The LEF file will be generated under the project directory path, open it and review it visually making sure that pins are listed there as expected and blockage layers and coordinates are the same as you placed.

```
-rw-rw-r--.  1 user01 user01 2.1K Jul 13 17:02 dig.lef
-rw-rw-r--.  1 user01 user01  798 Jul 13 17:02 lefout.log
[user01@centos7 proj1_tsmc65]$ nedit dig.lef&
```

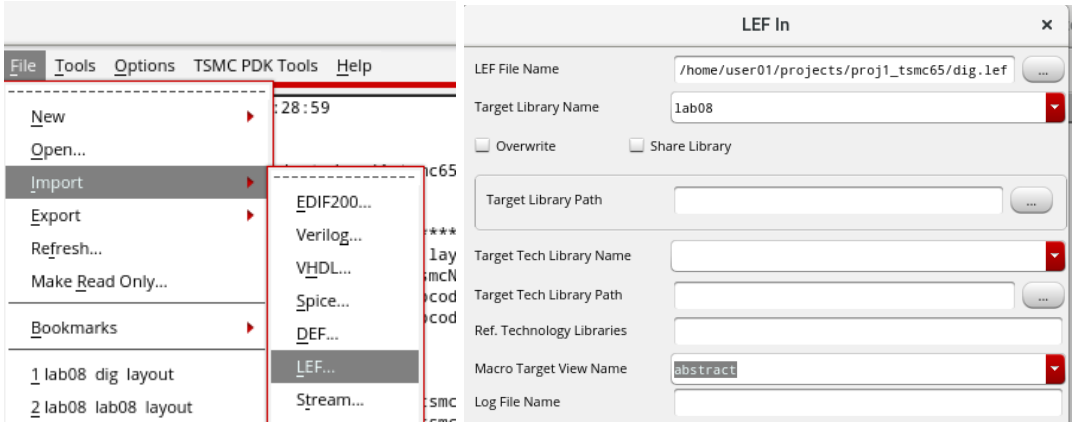


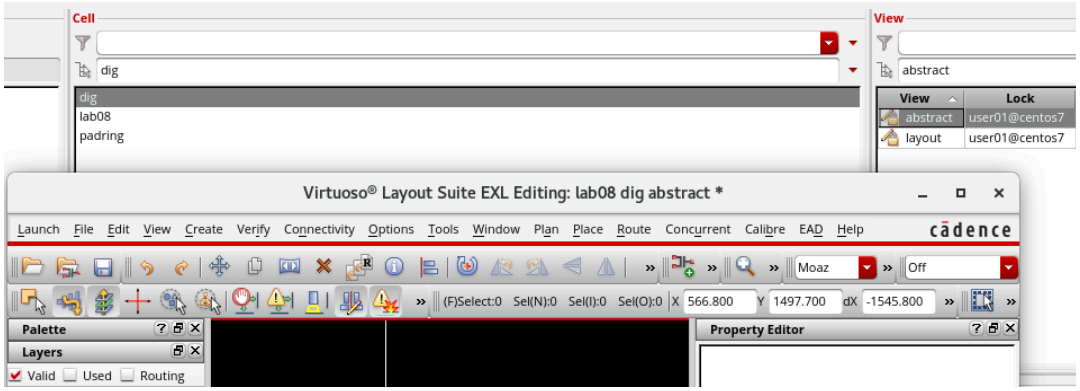
4. For more information about the LEF and DEF, read [the manual](#).

5. Similarly you can export the DEF file.



6. Import the LEF file as an abstract view to review it versus the layout view that you have created





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Lab Report

Prepare a PDF document in which you should include all the info/data required.
Explicit attachments should be uploaded to a drive and you should provide **unrestricted drive links** in the PDF report.

Lab08 library & PEX rule file download and installation

1. Download the “lab08” library & PEX rule file from here
2. Extract it under the VM shared directory
3. From the VM terminal, run the following commands:
   >> cd ~/projects/<proj-name>/
   >> cp -rf <path-to-shared-folder>/lab08 .
   >> cp -rf <path-to-shared-folder>/1p9m_6X1Z1U .
4. Open Virtuoso and check the cells under “lab08” library.
   If Virtuoso is already opened, then just goto **View > Refresh**.

Task1: Designing the PAD

1. Modify the wirebond rule file at the following path by copying it to your project workarea and do the following edits
   *$PDKDIR/Calibre/drc/wire_bond/CN65_WIRE_BOND_**9M_6X1Z1U**.20a*
   a. Open the single inline PAD with the 60um pitch switch
   b. Close the “*_STAGGER” switches

```
12
13 //OPTIONS
14 //=======
15
16 //#DEFINE PITCH_40_SINGLE        //Turn on to use IO cell (>=40um pitch) of the single_in_line pad ONLY
17 //#DEFINE PITCH_45_SINGLE        //Turn on to use IO cell (>=45um pitch) of the single_in_line pad ONLY
18 //#DEFINE PITCH_50_SINGLE        //Turn on to use IO cell (>=50um pitch) of the single_in_line pad ONLY
19 //#DEFINE PITCH_55_SINGLE        //Turn on to use IO cell (>=55um pitch) of the single_in_line pad ONLY
20 #DEFINE PITCH_60_SINGLE        //Turn on to use IO cell (>=60um pitch) of the single_in_line pad ONLY
21
22 //#DEFINE PITCH_45_STAGGER        //Turn on to use IO cell (>=45um pitch) of the staggered pad ONLY
23 //#DEFINE PITCH_50_STAGGER        //Turn on to use IO cell (>=50um pitch) of the staggered pad ONLY
24 //#DEFINE PITCH_60_STAGGER        //Turn on to use IO cell (>=60um pitch) of the staggered pad ONLY
25 //#DEFINE PITCH_70_STAGGER        //Turn on to use IO cell (>=70um pitch) of the staggered pad ONLY
26 //#DEFINE PITCH_80_STAGGER        //Turn on to use IO cell (>=80um pitch) of the staggered pad ONLY
27
28 //#DEFINE PITCH_80_TRITIER        //Turn on to use IO cell (>=80um pitch) of the tri_tier pad ONLY
29 /* Please note that only one kind of pitch can exist for each pad structure (single_in_line, stagger, tri_tier) */
30
31
```

2. Open Calibre > DRC, set the rule file path to the modified Wirebond (WB) rule file in the previous step and start running over the pad cell once, and then over the padring cells. They can be found under **lab08/PAD** and **lab08/padring** respectively.

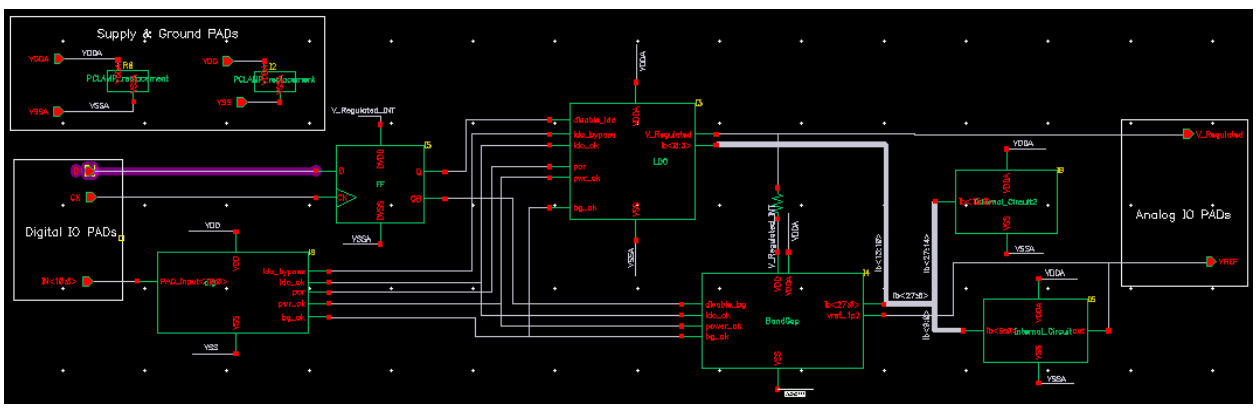
3. Solve the DRCs related to the drawn layers (CB,CB2,AP) and ignore the DRCs related to other metal layers.

4. Please add the following in the lab report
   a. The modifications done in the rule file
   b. Screenshots for the triggered WB violations in **lab08/PAD** and **lab08/padring**.

c. Screenshots for **lab08/PAD** and **lab08/padring** layouts after solving the violations with an illustration and annotation of the solution steps highlighting the PAD's layers final dimensions and pitches.
d. Please upload the following files to your drive and add their unrestricted access links to the PDF report
   i. DRC summary file before and after the wirebond fixes, it can be found under the following path */home/<user-name>/verification/proj1_tsmc65/tsmcN65/drc/<cellname>.drc.summary*
   ii. The modified WB rule file used in the run

Task2: Floorplan and Power Routing

It is required to floorplan the chip based on lab08/lab08 schematic and the below givens, constraints, and requirements



Givens

1. The schematics for the chip level and its all subsystems under lab08 library. This includes lab08, LDO, BG, Pclamp_replacement, Dig, mini-project1, Internal_circuit, Div2_FF, and Internal_circuit2.
2. The layouts for some blocks are available under lab08 library; PAD, Padring, Pclamp_replacement, LDO, Div2_FF, and BG.
3. All the pins inside the lab08 schematic are representing I/O PADs, so it is required to place AP pins on all pads on **lab08/lab08/layout** as if you are planning your padframe. i.e., pads order and location should be customized to be consistent with your adopted floorplan.
4. The metal stack is **9M_6X1Z1U**, so M9 should be used to connect between subsystems and PADs which are designed using AP RDL.
5. The LDO substrate is connected to **VSSA**.
6. The LDO connection to **V_Rgulated** PAD is handling very high current and hence requires very small parasitic resistance in routing.
7. The **VREF** PAD connection to the BandGap should be of small parasitic capacitance.

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Requirements

1. Do the floorplan for the top-level **lab08/lab08/schematic** while placing **lab08/padring** which won't be placed by generating all from source as it is not there in the schematic.

2. Create layout views with the pins **having metals;drawing under the pins (for the sake of LVS BOX which will come later)** with a **robust power plan** for each of the following subsystems similar to the LDO & BandGap layout views.

Block	Area (um²)
Dig	480,000
LDO	Layout view available under lab08 library
BG	Layout view available under lab08 library
Div2_FF	Layout view available under lab08 library
InternalCircuit	250,000
InternalCircuit2	250,000

3. Preserve and follow the manhattan used in the LDO and BandGap layout views. I.e., don't follow the manhattan used in the example illustrated at the beginning of the lab.

4. Implement the analog and digital supply rings, they should connect between blocks and PADs.

5. After allocating all pins on the pads, you can allocate multiple **VSSA, VDDA, VSS, and VDD pads** for better robustness (EMIR).

6. While allocating the pins over the pads, make sure to have the worst case resistance to the clamps "Pclamp_replacement" **≤ 1 ohm** as illustrated in the lecture.
   a. There are three methods to calculate the resistance from one point to another
      i. PERC (not available)

      ii. PEX point-to-point (P2P). Please refer to Appendix D for more info.
         1. You should be passing LVS or at least for the routes whose R being calculated
         2. Please use the rule file installed earlier at the following location */home/user01/projects/proj1_tsmc65/1p9m_6X1Z1U/calibre.rcx*
         3. It will be performed on two steps
            a. Once using the AP layer of the GND pad in the **From** layer and M7 of the "Pclamp_replacement" in the **To** layer
            b. Another time using the AP layer of the SPLY pad in the **From** layer and M7 of the "Pclamp_replacement" in the **To** layer

iii. If you don't want to use the PEX P2P, you can calculate the resistance manually using the resistance table in Appendix E. Please note that manual calculations will require further illustrations than point to point resistance

7. Add psub-rings around different blocks such that each ring is connected to the proper domain.

8. It is required to pass LVS & ERC for the toplevel after fulfilling all the above requirements using the following rule file after copying and doing some edits to it. *$PDKDIR/Calibre/online/lvs_online/1p9m_6X1Z1U/calibre.lvs*
   a. Use LVS Box for all the blocks except Div2_FF block. Please refer to Appendix A for more details about "LVS Box" usage.
   b. Don't use the "Connect by name" option.
   c. Enter the supplies and ground names

9. **[Bonus 2 marks]** Having controlled DRCs is a plus using the file and switches illustrated in Task 4

Finally for the lab report, please add the following data with illustrations and justifications:

1. Screenshots showing and illustrating the adopted floorplan
2. The chosen pad order with illustrations showing how you did meet the resistance requirements for the assumed ESD structure.
   a. Please provide clear screenshots or write them down in an excel sheet then copy the cells to the pdf document and attach the excel sheet to your drive if this is feasible to you.
   b. Whether you choose manual calculations or point to point resistance, please provide screenshots and details
3. Screenshots of any edited options in the LVS GUI such as LVS Box.
4. Any additional layers added to pass LVS
5. Any challenges that you faced in order to pass LVS & ERC
6. Please upload the following files/folders to your drive and add their unrestricted access links to the PDF report
   a. lab08 library after finishing the whole task
   b. LVS report file after passing LVS & ERC, it can be found under the following path */home/<user-name>/verification/proj1_tsmc65/tsmcN65/lvs/<cellname>.lvs.report*
   c. ERC summary file after passing LVS & ERC, it can be found under the following path */home/<user-name>/verification/proj1_tsmc65/tsmcN65/lvs/<cellname>.erc.summary*
   d. **[Required only for the bonus]** DRC summary file after passing DRC, it can be found under the following path */home/<user-name>/verification/proj1_tsmc65/tsmcN65/drc/<cellname>.drc.summary*

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Task3: Create LEF Text File for the Dig block.

1. Create a LEF text file for the digital block layout that was created in task2.
2. Add OBS or routing blockage layer for metals above M5.
3. The digital block should only use M1-M5 metals. Thus, the supplies are on M5 and signals are on M3&M4.
4. Follow the same manhattan used for other blocks.
5. Please add the following in the lab report
    a. Screenshot of the digital layout view showing pin distribution
    b. Copy LEF content to the report under the screenshot
    c. Add a drive URL for the generated LEF file

Task4: Solving Latch-up Violations **[Bonus Task | 2 Marks]**

Part I

In **lab08/LDO**, there is a device connected to “V_Regulated” pad, so it is required to run DRC and solve any related LUP violations.

1. Copy the following rule file to your workarea and modify it as follows
   $PDKDIR/Calibre/online/drc_online/1p9m_6X1Z1U/calibre.drc
    a. Close the “#DEFINE DISCONNECT_AFTER_RESISTOR” by adding // before it
    b. Open the “#DEFINE DEFINE_PAD_BY_TEXT” switch by removing the // before it
    c. Enter the PAD, Supply, and ground names appropriately
       Example: VARIABLE VDD_TEXT "VDD?" “VCC?”
    d. Close “#DEFINE CHECK_LOW_DENSITY” switch
    e. Close “#DEFINE WLCSP_SEALRING” switch

2. Call the DRC and point to the modified rule file making sure that the LUP checks are enabled.
   Hint: please check Lab 02 if you forgot how to check that.

3. Please add the following in the lab report
    a. The modifications done in the rule file
    b. A screenshot of the triggered LUP violations.
    c. A screenshot of the layout after solving the violations along with an illustration and annotation of the solution steps.
    d. Please upload the following files to your drive and add their unrestricted access links to the PDF report
        i. DRC summary file before and after the LUP fix, it can be found under the following path
           /home/<user-name>/verification/proj1_tsmc65/tsmcN65/drc/<cellname>.drc.summary
        ii. The rule file used in the DRC run

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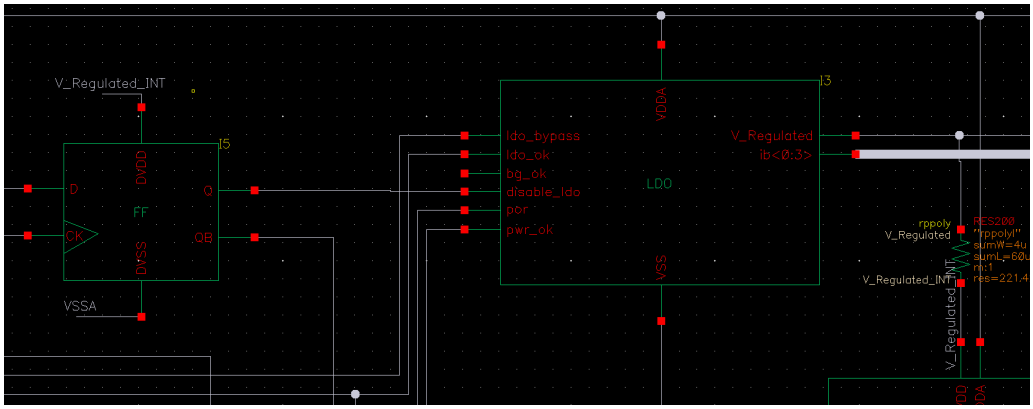
Part II

The V_Regulated_INT is a supply for the BandGap. Run DRC over lab08/lab08/layout once with “#DEFINE DISCONNECT_AFTER_RESISTOR” closed and another time having it closed.
And in the lab report, please add the following info

1. Show the LUP violations when the switch is closed and when it is opened
2. Write a justification for the reason behind that differences.



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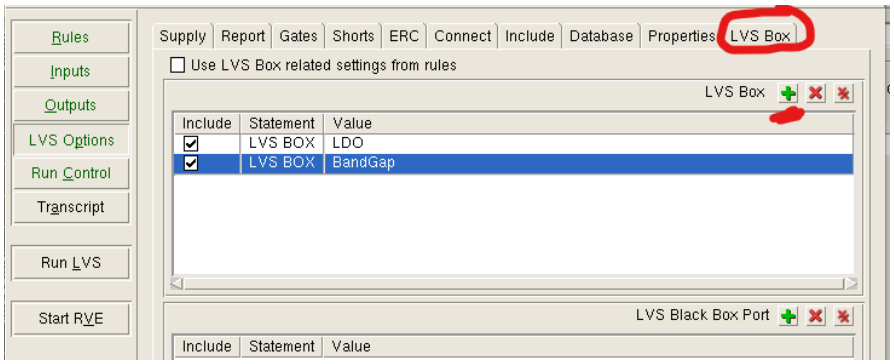
Appendix

A. How to run LVS Box?

From "LVS Options", goto "LVS Box" tab and do the following:

- Uncheck "Use LVS Box related settings from rules"
- Add all the blocks you need to ignore their internal connectivity as shown below

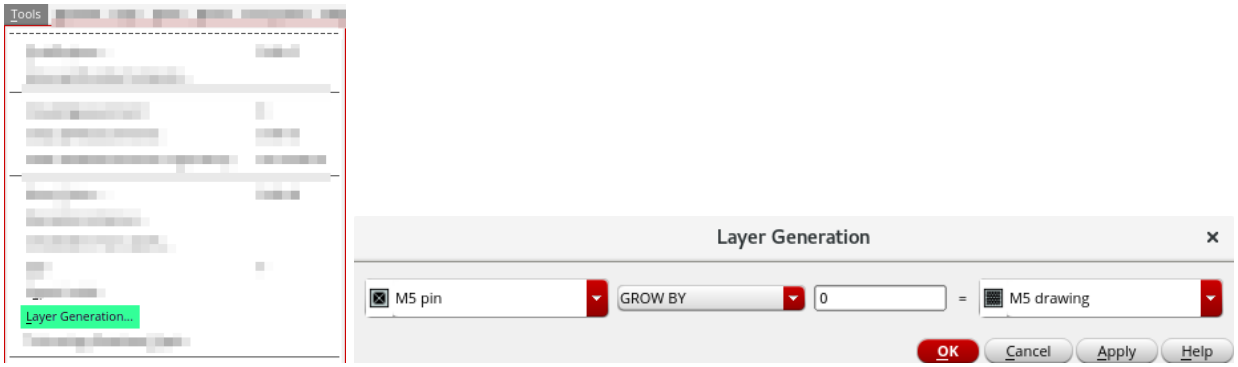


Please note that LVS Box won't work properly in the following cases, so you have to avoid these cases:

1. The blocks pins do not have metals' drawing layers underneath them. Please refer to Appendix B in order to solve it if you have any related issues
2. There are duplicate pins in the layout view of the block versus single pins in the schematic. This is usually intentional, so the solution here is to make sure that you short all these duplicate pins on the toplevel based on their connectivity requirements/constraints, i.e., they might be start connected to the PAD but at the end they are shorted.

B. How to auto generate "metal;drawing" under "metal;pin"?

From the layout, goto **Tools > Layer Generation…**
Select all pins or objects and apply the following generation on all the pin layers you have in the layout one by one. Each single generation is accomplished by clicking on "Apply"



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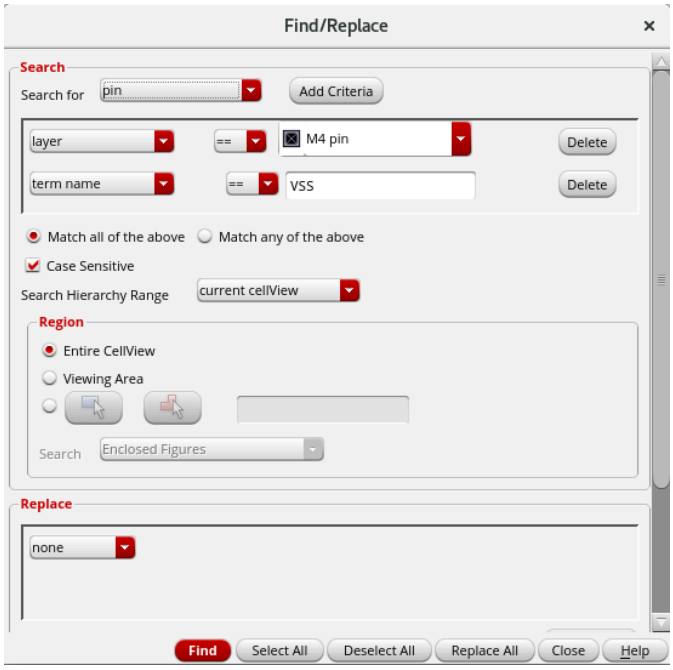
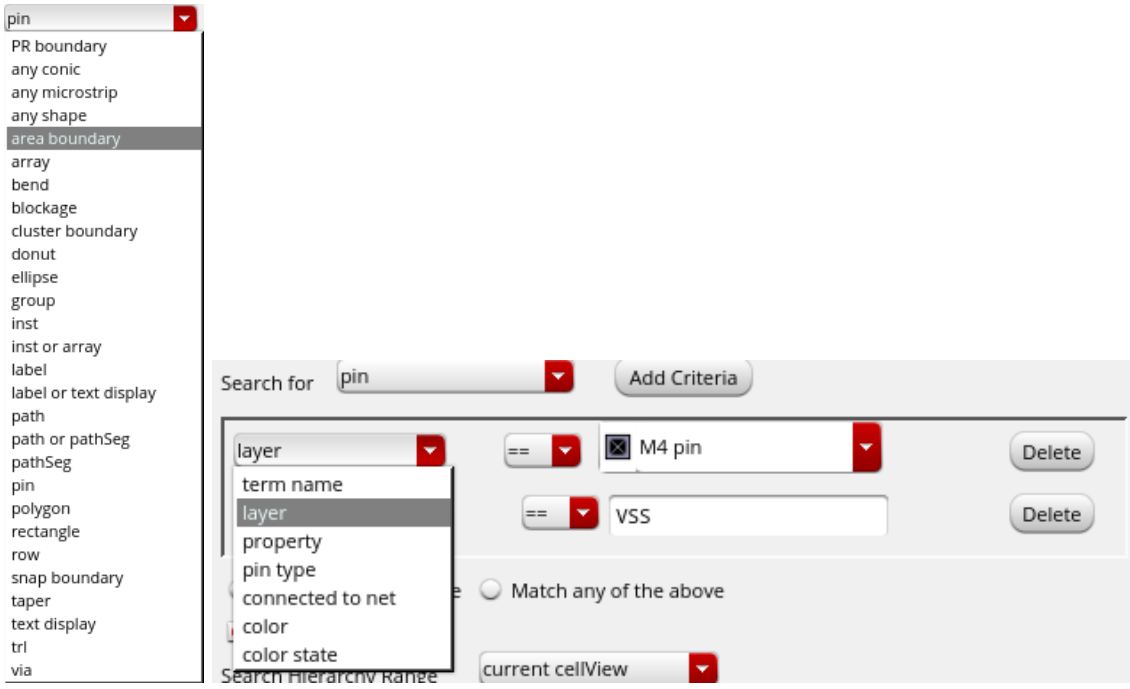
C. How to search for objects in the layout?

There are usually powerful search tools provided by the EDA vendor given the huge number of objects in the layout.
To call the search engine, press “Shift+s” where you can search objects and replace them easily



You can search for any object type and apply several filtration criteria



It is a very cool feature that worth the trial and usage

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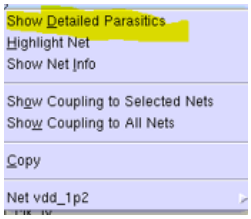
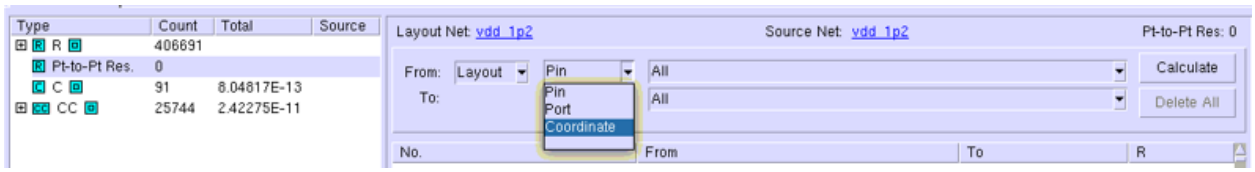
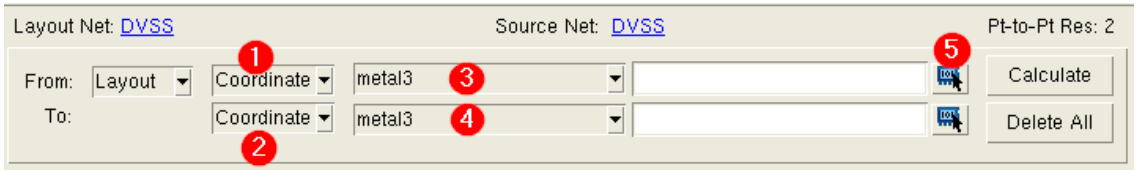
D. How to run Point-to-Point Resistance?

- Run PEX on the layout after passing LVS or at least making sure that you are not having any issues with routing that you are going to calculate its resistance **(I believe the tool more than myself in LVS)**.
- Please refer to the verification lecture or to mini-project1 for further details about running PEX.
- In the resultant parasitic tab, right click on any net and select “Show Detailed Parasitics”

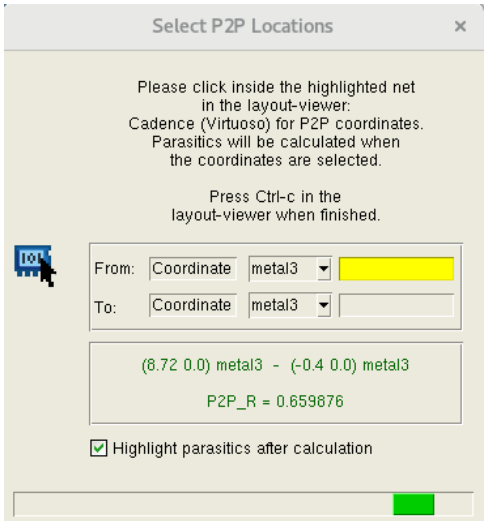


- Select “Pt-to-Pt Res” then on the right subwindow choose From “layout” “Coordinate” To “Coordinate” and choose the proper layers for the **From** and the **To,** then press on the button under 5





- A popup window will appear asking you to click on a point coordinate in the layout representing the source and then asks you to enter a second point which is the destination



- After that you will get the resistance calculated as shown

2	Coord 8.72 0.0 metal3	Coord -0.4 0.0 metal3	0.659876

E. Metals & VIAs Sheet Resistances

Layer	Sheet Resistance	Current Capability
M1	0.137 ohm/square	Please refer to the DRM
M2	0.119 ohm/square	Please refer to the DRM
M3	0.119 ohm/square	Please refer to the DRM
M4	0.119 ohm/square	Please refer to the DRM
M5	0.119 ohm/square	Please refer to the DRM
M6	0.119 ohm/square	Please refer to the DRM
M7	0.119 ohm/square	Please refer to the DRM
M8	0.0227 ohm/square	Please refer to the DRM
M9	0.005 ohm/square	Please refer to the DRM
V1	1.5 ohm/via	Please refer to the DRM
V2	1.5 ohm/via	Please refer to the DRM
V3	1.5 ohm/via	Please refer to the DRM
V4	1.5 ohm/via	Please refer to the DRM
V5	1.5 ohm/via	Please refer to the DRM
V6	1.5 ohm/via	Please refer to the DRM
V7	1.5 ohm/via	Please refer to the DRM
V8	0.22 ohm/via	Please refer to the DRM
V9	0.041 ohm/via	Please refer to the DRM